

Problem 3.31 has the uncertainty inequality facing the wrong way

A full counterexample to a problem in arXiv:2107.09057

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Status. Full counterexample, high confidence. Problem 3.31 as printed is false even for the group-subfactor planar algebra of $\mathbb{Z}_2$. Its premise is automatically satisfied by every normalized 2-box. The likely intended near-minimizer condition has the opposite inequality sign.

1 Source problem

The source is Wu, *Quantum smooth uncertainty principles for von Neumann bi-algebras* [1]. Problem 3.31 on source PDF page 15 asks for a positive function $C(\epsilon, \delta) \rightarrow 0$ as $\epsilon \rightarrow 0$ such that

$$H(|x|^2) + H(|\mathcal{F}(x)|^2) \geq 2 \log \delta - \epsilon \quad (1)$$

forces every normalized 2-box x of every irreducible subfactor planar algebra of Jones index δ^2 to lie within $C(\epsilon, \delta)$ of a bi-shift of a biprojection:

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- (1) In Example 2.5 take $\epsilon = \eta = 0$, then Theorem 3.28 implies Theorem 4.9 in 16;
 - (2) In Example 2.6 take $\epsilon = \eta = 0$, then Theorem 3.28 implies Theorem 5.5 in 9.

In 9, the minimizers of Hirschman-Beckner uncertainty principle on subfactor planar algebras were characterized as bi-shifts of biprojections (See Theorems 6.4 and 6.13 in 9). So it is natural to ask the following inverse problem.

Problem 3.31. Find a positive function $C(\epsilon, \delta)$, for $\epsilon, \delta > 0$, such that $\lim_{\epsilon \rightarrow 0} C(\epsilon, \delta) \rightarrow 0$, and for any 2-box x of any irreducible subfactor planar algebra with Jones index δ^2 , $\|x\|_2 = 1$, if

$$H(|x|^2) + H(|\mathcal{F}(x)|^2) \geq 2 \log \delta - \epsilon,$$

then $\|x - y\| \leq C(\epsilon, \delta)$ for some bi-shift of biprojection y .

4. AN ANSWER TO A CONJECTURE OF WIGDERSON AND WIGDERSON

The famous Heisenberg uncertainty principle in 6 could be mathematically for-

Immediately above the problem, the source recalls that bi-shifts are exactly the minimizers. More decisively, its Corollary 3.29 with the smoothing parameters set to zero gives

$$H(|x|^2) + H(|\mathcal{F}(x)|^2) \geq 2 \log \delta \quad (\|x\|_2 = 1). \quad (2)$$

Thus (1) does not express near-minimality: by (2), it holds for every normalized x and every $\epsilon > 0$.

2 Counterexample

Theorem 1. *No function C with the property required by Problem 3.31 exists. In the irreducible group-subfactor planar algebra associated with $G = \mathbb{Z}_2$, there is a normalized 2-box x satisfying the premise for every $\epsilon > 0$ whose distance from every bi-shift of a biprojection is at least $1/\sqrt{2}$ in the 2-norm and at least $1/2$ in the operator norm.*

3 Proof intuition

The group-subfactor example recovers ordinary finite Fourier analysis. On $\mathbb{Z}_2$, bi-shifts of subgroup indicators form only four complex lines: the two coordinate lines and the two Fourier-character lines. Choose a vector whose position and Fourier probability distributions are both uniform but whose relative phase keeps it off all four lines. It has entropy strictly *above* the minimum, yet the printed lower-bound premise accepts it for every error tolerance. A function tending to zero cannot bridge its fixed distance to the four minimizer lines.

4 Proof

Proof of Theorem 1. Jiang, Liu, and Wu explain that finite abelian group subfactors recover the classical Fourier uncertainty principle [2, Sections 2 and 8]. For $G = \mathbb{Z}_2$, identify the relevant 2-box with $\mathbb{C}^2$ in its orthonormal delta basis. The Jones index is 2, so $\delta = \sqrt{2}$, and the normalized Fourier transform is

$$\mathcal{F}(a, b) = \frac{1}{\sqrt{2}}(a + b, a - b).$$

Set

$$x = \frac{1}{\sqrt{2}}(1, i).$$

Then $\|x\|_2 = 1$, and

$$\mathcal{F}x = \frac{1}{2}(1 + i, 1 - i).$$

Consequently, both $|x|^2$ and $|\mathcal{F}x|^2$ are the uniform probability vector $(1/2, 1/2)$. With $H(p) = -\sum_j p_j \log p_j$,

$$H(|x|^2) + H(|\mathcal{F}x|^2) = 2 \log 2. \tag{3}$$

Since $2 \log \delta = \log 2$, equation (3) implies (1) for every $\epsilon > 0$.

It remains to quantify the failure of the conclusion. In a finite group subfactor, every bi-shift has the form

$$c \sum_{h \in K} \chi(h) \delta_{hg},$$

where $K \leq G$, $g \in G$, χ is a one-dimensional representation of K , and $c \in \mathbb{C}$ [2, Proposition 8.1]. The only subgroups of $\mathbb{Z}_2$ are $\{0\}$ and $\mathbb{Z}_2$. Therefore the set of all bi-shifts is the union of the four complex lines

$$\text{span}(1, 0), \quad \text{span}(0, 1), \quad \text{span}(1, 1), \quad \text{span}(1, -1). \tag{4}$$

For a unit vector u spanning any line in (4), direct calculation gives $|\langle x, u \rangle| = 1/\sqrt{2}$. Hence

$$\inf_{y \text{ a bi-shift}} \|x - y\|_2 = \sqrt{1 - \frac{1}{2}} = \frac{1}{\sqrt{2}}.$$

If the unadorned norm in Problem 3.31 denotes the operator norm, the same four-line calculation in ℓ_2^∞ gives the lower bound $1/2$: it is $1/\sqrt{2}$ for either coordinate line, while the smallest disk containing the two target coordinates for either character line has radius $1/2$.

Now suppose the requested C existed. Since $C(\epsilon, \sqrt{2}) \rightarrow 0$, choose $\epsilon > 0$ with $C(\epsilon, \sqrt{2}) < 1/2$. The vector x satisfies the premise, but no bi-shift satisfies the required conclusion in either natural norm. This is a contradiction. $\square$

5 Correction and scope

The standard stability question for a lower-bound inequality asks that the quantity be at most the minimum plus a small error. The likely intended hypothesis is

$$H(|x|^2) + H(|\mathcal{F}(x)|^2) \leq 2 \log \delta + \epsilon. \quad (5)$$

The present counterexample does not address the genuinely interesting uniform stability problem under (5); it refutes only Problem 3.31 exactly as printed.

The source also asks in Problem 4.7 for a sharp Fourier-norm constant on $\mathbb{R}$. A separate deep attempt checked Gaussian tests, equality conditions, comb limits, a Rényi reformulation, and the Euler–Lagrange equation. The Gaussian does not attain the source lower bound because the sharp Hausdorff–Young and interpolation equality conditions are incompatible. No full answer to Problem 4.7 is claimed here.

6 Verification and novelty

The counterexample was checked in two independent ways: first, the printed premise was compared directly with the source’s Corollary 3.29; second, all four $\mathbb{Z}_2$ bi-shift lines were enumerated using the finite-group classification and their exact distances from x were computed. No numerical approximation is used as proof.

The four lightweight run indexes had no entry for arXiv:2107.09057 or this sign obstruction. Bounded primary-source searches on 13 August 2026 used the exact Problem 3.31 wording, its entropy inequality, and stability of bi-shifts of biprojections. They found no correction or later treatment.

Human-review recommendation. High priority but quick: verify the classical $\mathbb{Z}_2$ normalization and confirm from the typeset source that the sign is indeed “ $\geq$ ”. The logical obstruction already follows from Corollary 3.29, independently of the explicit distance calculation.

References

- [1] J. Wu, *Quantum smooth uncertainty principles for von Neumann bi-algebras*, arXiv:2107.09057 (2021), Problem 3.31.
- [2] C. Jiang, Z. Liu, and J. Wu, *Noncommutative uncertainty principles*, J. Funct. Anal. 270 (2016), 264–311; arXiv:1408.1165.